

Q. Suppose the demand and supply of medical care are summarized by the following:

$$Q_d = 100 - 2P$$

$$Q_s = 20 + 2P$$

a) Draw the demand and supply curve in a same diagram.

b) Calculate equilibrium price and quantity

c) Suppose, health insurance is available and provides 20% coinsurance rate (hint: demand curve will shift); calculate new equilibrium price and quantity. (Show it in the same diagram).

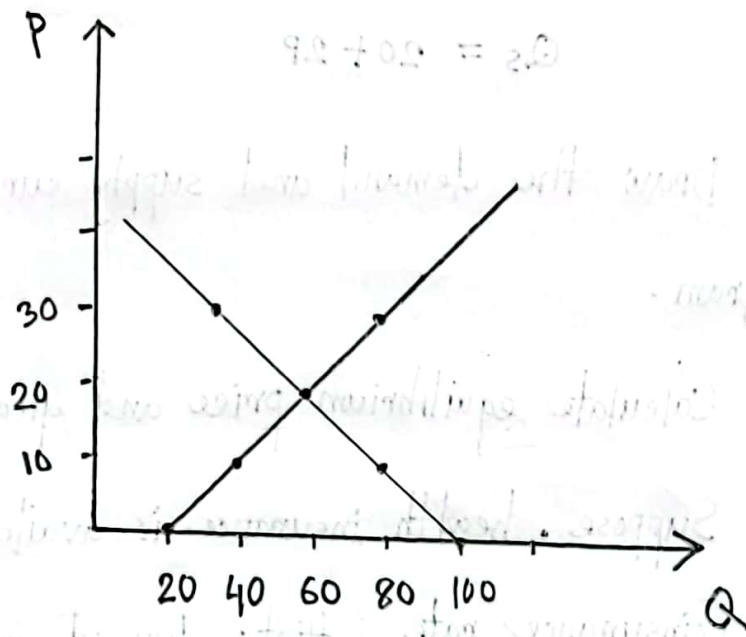
d) Calculate the deadweight loss.

e) Calculate the deadweight loss if coinsurance is 50%. Interpret your result.

90	9
80	10
70	20
60	30

100	0
90	10
80	20
70	30

Answer: to the question no: (a)



Given,

$$Q_d = 100 - 2P$$

$$Q_s = 20 + 2P$$

P	Q_d
0	100
10	80
20	60
30	40

P	Q_s
0	20
10	40
20	60
30	80

Answer to the question no: (b)

Here,

$$Q_d = 100 - 2P$$

$$Q_s = 20 + 2P$$

At equilibrium,

$$Q_d = Q_s$$

$$\Rightarrow 100 - 2P = 20 + 2P$$

$$\Rightarrow 4P = 100 - 20$$

$$\Rightarrow 4P = 80$$

$$\Rightarrow P = \frac{80}{4}$$

$$\therefore P = 20$$

Now, $Q_d = 100 - 2P$

$$= 100 - (2 \times 20)$$

$$= 100 - 40$$

$$= 60$$

Same as, $Q_s = 60$

So, $E(20, 60)$

Answer to the question no: (c)

Given, $Q_d = 100 - 2P$

$$\Rightarrow 2P = 100 - Q$$

$$\Rightarrow P = 50 - \frac{1}{2}Q \dots (i)$$

Now, given 20% coinsurance rate, then new demand equation can be found by dividing the equation (i) by 0.2; we get

$$P_d^* = (50 - 0.5Q)/0.2$$
$$= 250 - 2.5Q \dots (ii)$$

Again, $Q_s = 20 + 2P$

$$\Rightarrow 2P = Q_s - 20$$

$$\Rightarrow P = 0.5Q - 10 \dots (iii)$$

Now, new equilibrium Q and P can be found by equating (ii) and (iii),

$$P_d = P_s$$

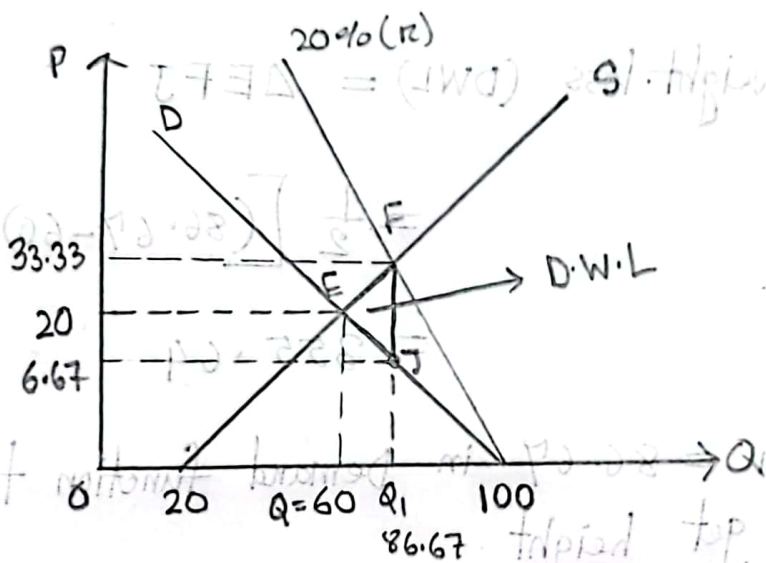
$$\Rightarrow 250 - 2.5Q = 0.5Q - 10$$

$$\Rightarrow 250 + 10 = 0.5Q + 2.5Q$$

$$\Rightarrow 3Q = 260$$

$$\therefore Q = 86.67$$

$$\begin{aligned} \text{Now, } P &= (0.5 \times 86.67) - 10 \\ &= 33.33 \end{aligned}$$



Therefore, new equilibrium price and quantity equal to 33.33 and 86.67

Answer to the question no: (d)

$$\text{Incremental cost} = \text{Area } QEFQ_1$$

$$\text{Incremental benefit} = \text{Area } QEJQ_1$$

$$\text{Therefore deadweight loss (DWL)} = \Delta EFJ$$

$$= \frac{1}{2} [(86.67 - 60) \times 6.67]$$

$$= 355.64$$

[Putting $Q_1 = 86.67$ in Demand function to get height.]

Profit is maximized when marginal revenue is equal to marginal cost.

Answer to the question no: (e)

$$P_d = P_s + c$$

$$\Rightarrow 50 - 0.5Q = (0.5Q - 10) + 0.5$$

$$\Rightarrow 50 - 0.5Q = 0.5Q - 9.5$$

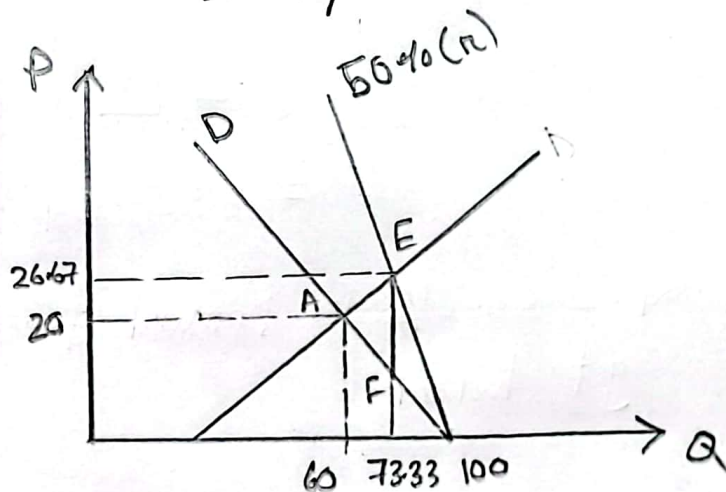
$$\Rightarrow 0.5Q + 0.5Q = 50 + 9.5$$

$$\Rightarrow 0.75Q = 55$$

$$\therefore \bar{Q} = \frac{55}{0.75} = 73.33$$

$$\therefore \bar{P} = 0.5(73.33) - 10$$

$$= 26.67$$



$$\therefore DWL = \Delta AEF$$

$$= \frac{1}{2} (73.33 - 60) \times (26.67 - 13.34)$$

$$= \frac{1}{2} \times 13.33 \times 13.33$$

$$= 88.845$$

Interpretation:

When the coinsurance rate is 20%, then the deadweight loss 355.64 unit. And quantity demanded is 86.67 and price will be 33.33.

Again when the coinsurance rate is 50%, then the quantity demanded is 73.33 unit and the deadweight loss is 88.845.

So, it can be said that, with the increase of coinsurance rate, the deadweight loss will be decreased. i.e. there is an inverse relationship between coinsurance rate and deadweight loss.

$$(100.01 - 73.33) \times (22 - 33.33) \times \frac{1}{2} =$$

$$88.845 \times 10.01 \times \frac{1}{2} =$$

$$444.22 =$$